# University Student-Support Case Agent

BSE4104 Emerging Trends in Software Engineering

Group M Evening | Final Presentation

- **Grounded answers**
- **Bounded actions**
- **Human hand-off**

# The problem and who it is for

## User

Makerere freshmen (first-year) going through onboarding

## Pain

Answers scattered across documents and offices; queues for routine FAQs

## Need

Grounded answers, plus a support draft when the documents are not enough

**Out of scope**

Admissions, grading, discipline, fee decisions and auto-resolving cases

# Why AI is justified, and where it stops

### AI

Retrieval-grounded Q&A, plus deciding when to escalate to a draft

### Deterministic software

Validation, case IDs, queue routing, limits and the consent gate

### Human staff

Review and resolve every draft

**AI does not invent policy, waive fees, close cases, or act without agreement.**

# From model to bounded agent in eight weeks

**W2 Foundation model**

Versioned prompts, v1.0 to v1.1

**W3 RAG**

Corpus, retrieve, cite, honest fallback (v1.2)

**W4 Tools**

retrieve_documents and create_support_case (v1.3)

**W5 Bounded agent**

Sense, Plan, Act, Observe, Stop

**W6 State and memory**

Explicit state, justified memory, MCP-style interface

**W7 Evaluation**

Observability, guardrails, failure catalogue

Week 8: hardened release, final report and demonstration

# The bounded agent loop

**Sense**

Blocked topic, non-freshman and injection checks

>

**Plan**

Model requests only approved tools

>

**Act**

Application runs allow-listed tools only

>

**Observe**

Tool status, then re-plan or stop

>

**Stop**

Conditions S1 to S8: grounded, draft, refuse, limits

re-plan, at most 3 rounds per turn

**Limits are enforced in application code, not by the model**

**3** model rounds per turn

**3** tool calls per turn

**1** support draft per session

# Live demo: from question to staff hand-off

1

### Ask a grounded question

Cited source, tool used: retrieve_documents, stop condition S1

2

### Ask for staff help

Agree in the sidebar and give a valid registration number. Case created: case id, queue label, stop condition S2

**The consent control lives in the sidebar. The case tool stays off until the student agrees.**

# We can inspect what the agent did

## What every turn records

| | |
|---|---|
| **Tool name** | which tool the model asked for |
| **Arguments** | what the call was made with |
| **Status** | ok, no_match, validation_error, created |
| **Stop condition** | S1 to S8, why the turn ended |
| **Sources** | which documents grounded the answer |

**16/16**

deterministic checks passed after the Week 8 fixes

### Evidence we can open

- Agent traces T1 to T3 and O1 to O4
- Guardrail screenshots G01 to G10
- Scenario evaluation results

# Failure and recovery: bad input is caught

1

**Student gives 23/A/93939**

>

2

validation_error

**The application rejects the format**

>

3

**The agent asks for a valid number (U or X, not A)**

>

4

draft created

**Retry with a valid number**

! **Also observed**

Gemini 503 or 429: the student sees a short busy sentence and can retry. The raw error stays in the log.

**Lesson**

The application owns validation. The model cannot invent a valid case.

# Guardrails and human control

### Agreement first

Consent is a sidebar control; the case tool stays off until the student agrees

**G01**

### Blocked topics

Grade changes and fee waivers stop at S5. A policy question is not blocked in code.

**S5 | G03, G04**

### Scope and injection

Non-freshman requests stop at S4; override attempts stop at S6

**S4, S6 | G05, G06**

### One draft per session

A second draft is refused and points to the existing case

**G08**

### Tool allow-list

Only retrieve_documents and create_support_case can run

**Two tools**

### Human hand-off

Every draft goes to staff; the agent never resolves or closes cases

**Staff resolve**

# Known limitations

### Pattern-based injection check

It catches the override phrasings we listed, not every adversarial wording

### Retrieval cutoff 0.70

Some fee and joining passages still miss. The agent refuses rather than guess.

### Shared free-tier API

A 503 or 429 can interrupt a session. The student sees a short busy sentence. The log keeps the raw error.

# Key lessons

1. Grounding and citations beat unrestricted chat
2. Orchestration must enforce limits; do not trust the model alone
3. Consent is a UI flag, not a free-text yes
4. Evaluate refusals and failures, not only happy paths
5. Keep scope small enough to test, explain and control

# Thank you.
# Questions?

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BSE4104 Capstone

## Repository and evidence

github.com/SumayyaN/BSE41014_capstone_group_m_eve

App: python -m streamlit run src/app.py

Traces: evidence/traces/

Guardrails: evidence/screenshots/guardrails/

Evaluation: docs/evaluation/

Reports: docs/weekly-reports/ and docs/Architecture/